

Bilguun Dugarsuren

Applied Mathematics and CS Undergraduate

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As an Applied Mathematics and CS Undergraduate, I excel in programming, AI, physical computing, and media production, with a focus on cross-cultural storytelling. My public speaking, leadership, and teamwork skills enhance my collaborative projects. My proficiency in Python and Large Language Models (LLM) boosts my AI application development and R&D experience.

Work Experience

Promotional material graphic designer & Outreach

Mar 2026 - Present

Humans of DKU | Duke Kunshan University

Led promotional and outreach efforts for a student-run publishing organization at DKU, including designing materials and the official booklet for the Spring 2026 Humans of DKU exhibition.

Resident assistant

Aug 2025 - Present

Duke Kunshan University Residence Life | Duke Kunshan University

- Developed and implemented engaging recreational programs to enhance residents' quality of life.
- Trained and supervised new staff members on best practices in resident care and facility policies.
- Individually and collaboratively organized 20+ community events within the first semester of Fall 2025 to promote resident participation and enjoyment.
- Currently in the progress of unofficial and individual developing inventory management system for ResLife Inventory

Sports Complex Associate

Sep 2024 - Present

DKU athletics | Duke Kunshan University

- Developed visually compelling designs for print and digital media that enhanced brand identity and engagement.
- Facilitated sports events and activities, coordinating schedules and resources to optimize participation and engagement.
- Implemented customer service best practices, resolving inquiries and issues promptly to enhance visitor satisfaction.
- Managed inventory and equipment, ensuring availability and readiness for various sports activities.
- Trained and mentored more than 10 new staff on operations and customer service protocols, fostering a knowledgeable and capable team.

Assistant English teacher

Jun 2022 - Jul 2025

AKOK Academy | Ulaanbaatar

- Evaluated over 200 student progress and provided constructive feedback to enhance language acquisition and comprehension.
- Developed engaging lesson plans that catered to diverse learning styles and fostered student participation.
- Incorporated technology tools such as interactive whiteboards and language learning software to facilitate lessons.

Projects

Radikal

Apr 2025 - Apr 2026

RADIKAL is an AI-powered Mandarin Chinese learning app that personalizes character memorization through AI-generated mnemonics and visual associations. I co-led front-end development, designing the full interface structure using HTML, Tailwind CSS, and TypeScript with iterative user testing, while also developing the back end by designing the database schema and authentication and user session management system in Supabase, building server-side logic in TypeScript, integrating APIs, engineering prompts for AI-generated content, and managing environment configuration across development and production.

Smart Financial Planner

Nov 2025 - Nov 2025

Developed a web-based financial planning tool that helps users track expenses, plan budgets, and visualize financial goals. Designed and implemented core features using web technologies, including user input forms, data processing logic, and interactive financial calculations. Integrated data visualization components to display spending patterns and

budget allocations, and contributed to structuring the application workflow to support clearer financial insights and decision-making. This project involved research & development into user financial behaviors.

Duke-DKU Presence Lab Bass Connections Project-Duke University & Duke Kunshan University
Student Researcher (Accepted)

Aug 2026 - Aug 2027

Portfolio on my journey through "Processing"

Oct 2025 - Dec 2025

Designed and developed a portfolio of computational art projects using Processing (Java), encompassing generative graphics, animations, and a fully functional 2D game. Each project explored distinct visual and algorithmic concepts, progressively challenging my creative and technical range. Compiled and presented the full body of work through a self-designed portfolio website built in Canva.

Core Skills

Programming languages: Python, Java, R

Application Development, AI, LLMs, R&D, Web design, Graphic design, AI application development, Physical computing, Media production, Public speaking, TEDx Talk, Data analysis, Computer graphics, Cross-functional collaboration

Education

Duke Kunshan University

2024 - 2026

Bachelor of Science (BS) Applied Mathematics and Computational Sciences- CS track

Mongol Aspiration International Laboratory School

2019 - Jun 2023

High School Diploma

Languages

Mongolian (Native), English (Fluent), Chinese (Intermediate), Japanese (Beginner)

Awards

Best Associate of the Year

2025

DKU athletics

Dean's list

2024

Vibe Coding Marathon-1st Place

Nov 2025

DKU Innovation Lab

Interests

Programming, Public speaking, Web design, Photography, Video Editing, Basketball, Learning